BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Combining Boolean Logic

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What are the NOT outputs for input 0 then input 1?

2. What does a structure diagram show?
   A) The order instructions run
   B) Parts and their hierarchy
   C) Every variable's value
   D) The network's IP addresses

3. "Computers use binary because humans cannot write numbers above 1." Correct this. Explain why.

4. A program reads exactly ten scores. Another asks for a password until a condition is met. Correct both claims. Explain why. (a) Count-controlled loops are only for unknown repeat counts. (b) Condition-controlled loops always run ten times.

Today's objectives

Read a compound Boolean expression. Evaluate intermediate gates in order.

Key vocabulary - Read, cover, recall

Combined logic
Several linked gates forming a Boolean expression.

Intermediate output
Result from a gate before the final gate.

AND
True only with two high inputs.

OR
True when at least one input is high.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Build gates in sequence
Combined logic sends one gate's output into another; label each intermediate wire clearly.

Work inside parentheses
For Q=(A AND B) OR C, calculate X=A AND B before finding Q=X OR C.

All three-input rows
With A,B,C, there are eight input combinations: 000,001,010,011,100,101,110,111.

Check intermediate values
Record X and Q separately to spot where a wrong gate evaluation changes the final result.

I Do - Worked model

A circuit is Q=(A AND B) OR C. Find X=A AND B and Q for inputs (1,0,1) and (1,1,0). [4 marks]

- Work from the AND gate, X=A AND B, then OR X with C.
- For inputs (1,0,1), X=1 AND 0=0 [1].
- Q=0 OR 1=1 [1].
- For (1,1,0), X=1 [1] and Q=1 OR 0=1 [1].

We Do - Guided practice

1. For Q=(A AND B) OR C, evaluate inputs 0,1,0 and 1,1,1. Show X and Q. [4 marks]

2. A different system uses Q=NOT A OR B. Find Q when inputs are 1,0 and 0,0. [4 marks]

3. Explain why Q=(A AND B) OR C is always 1 when C=1. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the condition so the alert appears when malware is found or when both login checks fail.

```
unknownDevice = true
repeatedFailures = false
malware = true
if ... then
print("Investigate")
endif
```

You Do - Independent exam questions

Q2. State the intermediate expression X for Q=(A AND B) OR C. [1 mark]

Q3. Find X and Q for (A,B,C)=(0,0,1) and (1,1,0). [4 marks]

Q4. Complete the Q column for Q=(A AND B) OR C with inputs 000,001,010,011,100,101,110,111, and explain the 110 output. [6 marks]

Exit Ticket - True or False?

For Q=(A AND B) OR C, Q=1 whenever C=1.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. 1 then 0.

2. B - Parts and their hierarchy

3. Digital circuits can reliably distinguish two states, represented by 0 and 1.

4. (a) Use a count-controlled loop when the number of repeats is known. (b) Use a condition-controlled loop when a condition decides when it stops. Choose the right loop and trace its steps.

I Do - Explain the model

Work from the AND gate, X=A AND B, then OR X with C.

For inputs (1,0,1), X=1 AND 0=0 [1].

Q=0 OR 1=1 [1].

For (1,1,0), X=1 [1] and Q=1 OR 0=1 [1].

We Do - Model answers

1. For 0,1,0: X=0 [1] and Q=0 [1]. For 1,1,1: X=1 [1] and Q=1 [1].

2. For 1,0, NOT A=0 [1] so Q=0 [1]. For 0,0, NOT A=1 [1] so Q=1 [1].

3. X=A AND B can be 0 or 1 [1]. OR combines X with C [1]. With C=1, at least one OR input is 1 [1], so Q=1 [1].

Code First - Answer

```
unknownDevice = true
repeatedFailures = false
malware = true
if (unknownDevice AND repeatedFailures) OR malware then
print("Investigate")
endif
```

Expected output: Investigate

You Do - Self-mark

Q2: X=A AND B [1].

Q3: For 0,0,1 X=0 [1], Q=1 [1]. For 1,1,0 X=1 [1], Q=1 [1].

Q4: Q column: 0,1,0,1,0,1,1,1 [4 marks: one for each correct adjacent pair]. For 110, A AND B=1 [1], so 1 OR C=0 gives Q=1 [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.